

Math 199, Spring 2026
Yigal Kamel
2/4/26

Week 3 - Sets and prime numbers

1) For each of the following sets, describe the set in words, write down how many elements it has, and give one example of a proposition (true or false) that is quantified by the set in question. The proposition can be about math or not. Write whether the proposition is true, false, or whether (and why) it is hard to determine.

(a) $\mathbb{Z}$

(b) $\mathbb{Q}$

(c) $\mathbb{R}$

(d) $\emptyset$

(e) $\{\emptyset, \{\emptyset\}, 0\}$

(f) $\{x \in \mathbb{R} : x^2 = 2\}$

(g) $\{x \in \mathbb{Q} : x^2 = 2\}$

(h) $\{\{\{5\}\}\}$

(i) $\{n \in \mathbb{Z} : 2n + 7 \text{ is a prime number}\}$ (see also problem (6))

(j) $\{x \in \mathbb{Q} : 0 < x < \pi \text{ and either } 2x \in \mathbb{Z} \text{ or } 3x \in \mathbb{Z}\}$

(k) $\{x \in J : 2x \text{ is prime or } 3x \text{ is prime}\}$, where J is the set from part (j).

2) Write down a definition for what it means for a positive integer to be a prime number.

3) Let P be the statement: For all $x, y \in \mathbb{R} \setminus \{1\}$, if xy is a prime number then x and y are not both integers. Either prove P is true or give a counterexample. (*Hint*: What is the contrapositive?)

4) Let Q be the statement: For all $x, y \in \mathbb{R} \setminus \{1\}$, if xy is a prime number then x and y are both not integers. Either prove Q is true or give a counterexample. (What is the difference between Q and P from the previous problem?)

5) Prove that for any two integers $p, q > 1$, the number $pq + 1$ is not divisible by p or q .

6) Prove that for every integer $x > 2$, the number $x^2 - 1$ is not prime.

5) Let $E_{30} = \{n \in \mathbb{Z} : 2 < n \leq 30\}$. Prove that every element of E_{30} can be written as a sum of two prime numbers.

Do you think that every even integer can be written as a sum of two prime numbers? Write down any ideas you might have for approaching this. This is an open problem called Goldbach's conjecture.

6) Prove that there are infinitely many prime numbers. *Hint:* Assume that there are finitely many prime numbers and derive a contradiction.